

EXPERIMENT NO 6

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Div:- SY3

Batch:- C3

INPUT CODE

```
#include <stdio.h>
#include <stdlib.h>

struct node {
    int data;
    struct node *next;
};

struct node *head = NULL;

void push(int val) {
    struct node *newNode = (struct node*)malloc(sizeof(struct node));
    if (newNode == NULL) {
        printf("Memory allocation failed!\n");
        return;
    }
    newNode->data = val;
    newNode->next = head;
    head = newNode;
    printf("%d pushed to stack.\n", val);
}

void pop() {
    struct node *temp;
    if (head == NULL) {
        printf("Stack is empty (Underflow)\n");
        return;
    }
    temp = head;
    head = head->next;
    printf("%d popped from stack.\n", temp->data);
    free(temp);
}

void printList() {
```

```

struct node *temp = head;
if (head == NULL) {
    printf("Stack is empty\n");
    return;
}
printf("Stack contents: ");
while (temp != NULL) {
    printf("%d -> ", temp->data);
    temp = temp->next;
}
printf("NULL\n");
}

```

```

int main() {
    int choice, value;

    while (1) {
        printf("\n--- Stack Menu ---\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Display Stack\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter the value to push: \n");
                scanf("%d", &value);
                push(value);
                break;
            case 2:
                pop();
                break;
            case 3:
                printList();
                break;
            case 4:
                printf("Exiting...\n");
                while (head != NULL) {
                    pop();
                }
                exit(0);
            default:

```

```

        printf("Invalid choice! Please try again.\n");
    }
}
return 0;
}

```

OUTPUT

```

--- Stack Operations Menu ---
1. INSERT (Push)
2. DELETE (Pop)
3. DISPLAY
4. Exit
Enter your choice: 1
Enter value to insert: 12
12 inserted into the stack.

```

```

--- Stack Operations Menu --
1. INSERT (Push)
2. DELETE (Pop)
3. DISPLAY
4. Exit
Enter your choice: 1
Enter value to insert: 45
45 inserted into the stack.

```

```

--- Stack Operations Menu ---
1. INSERT (Push)
2. DELETE (Pop)
3. DISPLAY
4. Exit
Enter your choice: 1
Enter value to insert: 56
56 inserted into the stack.

```

```

--- Stack Operations Menu ---
1. INSERT (Push)
2. DELETE (Pop)
3. DISPLAY
4. Exit
Enter your choice: 1
Enter value to insert: 890
890 inserted into the stack.

```